**Tutorial Sheet 2**

**IIT Delhi – Abu Dhabi**

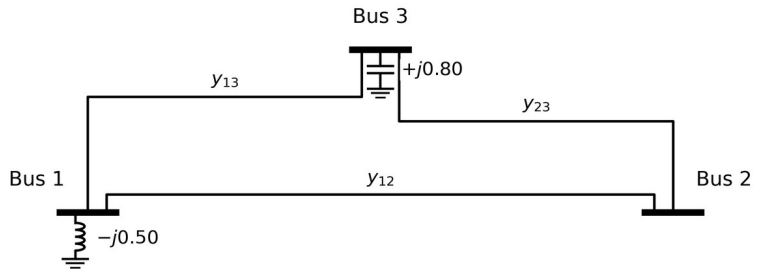
**$Y_{bus}$ Matrix Formation**

**Course Name: Power Electronics and Power Systems**

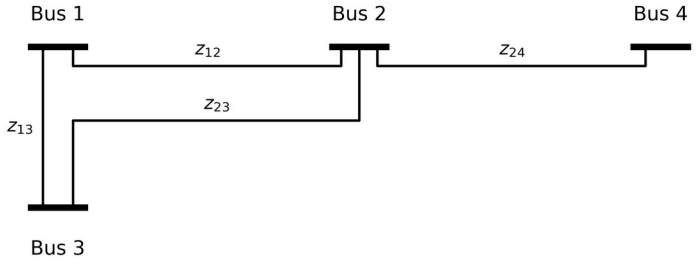
**Course Code: AENL226**

*Unless otherwise stated, all quantities are in per unit.*

1. A three-bus network has series admittances $y_{12} = 2 - j6$, $y_{13} = 1 - j4$, and $y_{23} = 3 - j8$. A shunt reactor $-j0.50$ is connected at bus 1, and a shunt capacitor $+j0.80$ is connected at bus 3. Form $Y_{bus}$ by inspection and verify each row sum.




2. The branch impedances are $z_{12} = 0.04 + j0.12$, $z_{13} = 0.05 + j0.15$, $z_{23} = 0.10 + j0.20$, and $z_{24} = 0.08 + j0.16$. Line-charging admittance and shunt elements are neglected. Form the $4 \times 4$ $Y_{bus}$ and explain why $Y_{34} = 0$.



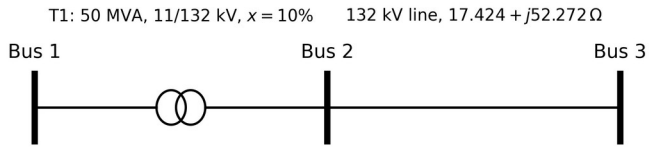

3. The transmission-line data for a five-bus power system are given in the following table. All quantities are in per unit. The parameters $r_{ij}$ and $x_{ij}$ represent the series resistance and reactance, while $b_{ij}$ represents the total line-charging admittance.

| Bus $i$ | Bus $j$ | $r_{ij}$ | $x_{ij}$ | $b_{ij}$ |
|---|---|---|---|---|
| 1 | 2 | 0.027 | 0.32 | 0.15 |
| 1 | 5 | 0.014 | 0.18 | 0.10 |
| 2 | 3 | 0.012 | 0.13 | 0.12 |

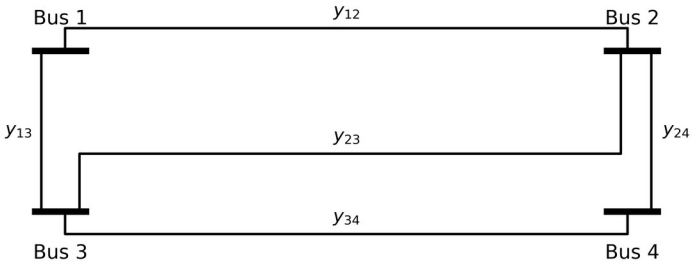
| Bus $i$ | Bus $j$ | $r_{ij}$ | $x_{ij}$ | $b_{ij}$ |
|---|---|---|---|---|
| 3 | 4 | 0.025 | 0.25 | 0.00 |
| 3 | 5 | 0.017 | 0.20 | 0.08 |

Using the nominal-$\pi$ model for each transmission line, form the $5 \times 5$ bus-admittance matrix $Y_{bus}$. Express the final matrix in rectangular form.

4. A 50 MVA, 11/132 kV transformer connects bus 1 to bus 2 and has 10% leakage reactance on its own rating. A 132 kV transmission line connects bus 2 to bus 3 and has per-phase impedance $17.424 + j\,52.272\,\Omega$. Using a 100 MVA system base, with voltage bases of 11 kV at bus 1 and 132 kV at buses 2 and 3, convert all series impedances to per unit and form the $3 \times 3$ $Y_{bus}$. Neglect line-charging admittance and shunt elements.




5. A four-bus network has $y_{12} = -j5$, $y_{13} = -j4$, $y_{23} = -j2$, $y_{24} = -j3$, and $y_{34} = -j6$, with no shunts. Form $Y_{bus}$ and show that it is singular.




6. A four-bus system contains a 40 MVA, 11/66 kV transformer between buses 1 and 2 with 8% leakage reactance; a 66 kV transmission line between buses 2 and 3 with per-phase impedance $4.356 + j\,17.424\,\Omega$ and total line-charging admittance $j\,0.04$ pu on the common 50 MVA, 66 kV base; and a

25 MVA, 66/22 kV transformer between buses 3 and 4 with 10% leakage reactance. On a common 50 MVA base, with voltage bases following the transformer ratios, convert the transformer and line impedances to per unit and form the $4 \times 4$ $Y_{bus}$. Why is a $Y_{bus}$ matrix sparse?

